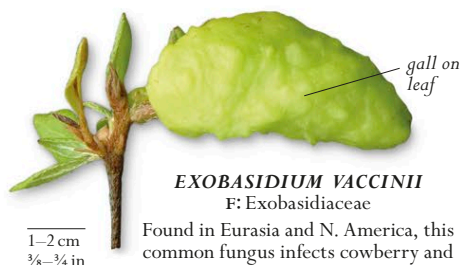


EXOBASIDIALES

This small group consists mainly of gall-forming plant parasites whose spore-producing cells form a layer on the leaf surface. Some cause disease in cultivated plants of the *Vaccinium* genus, which includes the common blueberry.



EXOBASIDIUM VACCINII

F: Exobasidiaceae

Found in Eurasia and N. America, this common fungus infects cowberry and eventually turns the leaves bright red. The leaves may become distorted and gall-like.

1–2 cm
3/8–3/4 in

UROCYSTIDIALES

This order contains some well-known smut fungi, in particular species of the genus *Urocystis*. These are parasites of flowering plants such as anemone, onions, wheat, and rye, and often cause serious injury to the host plant.

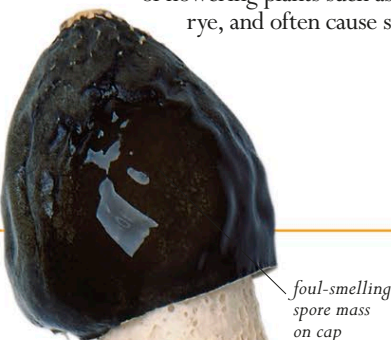
ANEMONE SMUT

Urocystis anemones

F: Urocystidaceae

This smut fungus from Eurasia and N. America forms dark brown, powdery, raised pustules on the leaves of anemones and some other plants.

2–4 mm
1/16–1/32 in



foul-smelling
spore mass
on cap

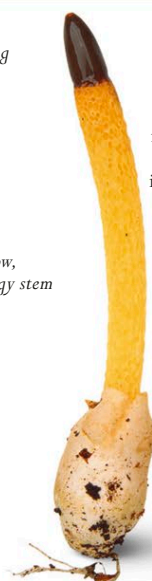
DOG STINKHORN

Mutinus caninus

F: Phallaceae

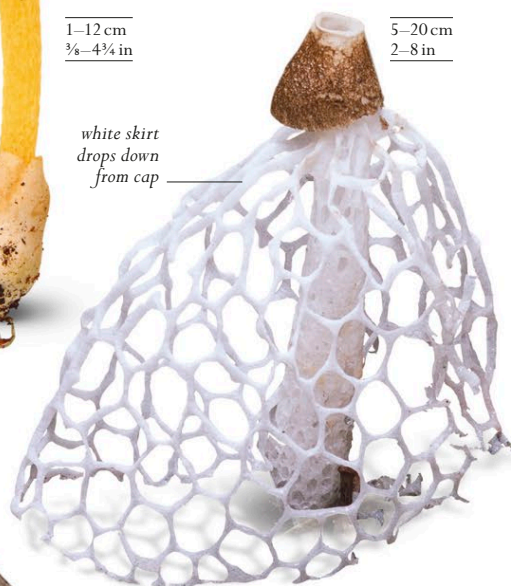
This common stinkhorn is found in mixed woods in Eurasia and N. America. Its tip, covered in greenish black spores, is joined to the spongy stem that emerges from a white "egg".

1–12 cm
3/8–4 3/4 in



hollow,
spongy stem

white skirt
drops down
from cap



PHALLUS MERULINUS

F: Phallaceae

This tropical species is mainly found in Australasia. It hatches from a white "egg". There are many similar species, some with brightly coloured "skirts".

large,
white "egg"

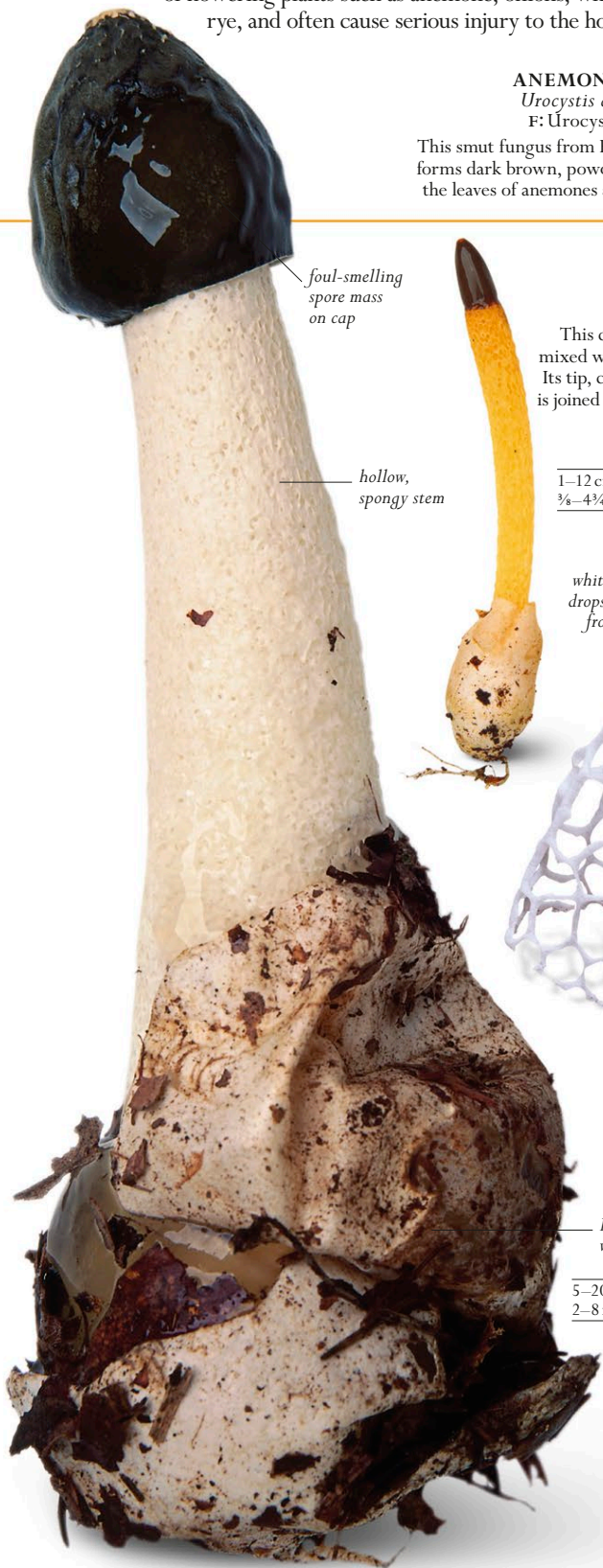
5–20 cm
2–8 in

STINKHORN

Phallus impudicus

F: Phallaceae

Common in mixed woods, this fungus is found in Eurasia. Its spore-covered, honeycomb-like cap hatches in a few hours from an "egg". Its foul odour is often evident many metres away.



PUCCINIALES

One of the largest orders of fungi with over 7,000 species, the rust fungi include numerous serious parasites of crop plants. They can have very complex life cycles with multiple hosts and produce different types of spores at different stages in their life.

black, powdery spores
in yellow spots on
lower leaf surface



RASPBERRY YELLOW RUST

Phragmidium rubi-idaei

F: Phragmidiaceae

This rust from Eurasia and N. America causes pustules to form on the upper surface of leaves. It survives winter with the help of black spores on the underside of leaves.

orange rust
damaging
rose stem



ROSE RUST

Phragmidium tuberculatum

F: Phragmidiaceae

This common rust is found in N. America and Eurasia. It causes orange pustules on the undersides of leaves and on distorted stems. The pustules turn black in late summer.

ALEXANDERS RUST

Puccinia smyrnii

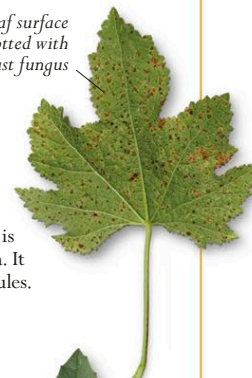
F: Pucciniaceae

Found across Eurasia, this common rust fungus forms raised plaques or warts on the leaves of Alexanders (*Smyrniolus satrum*).



yellow warts
of rust fungus

leaf surface
spotted with
rust fungus



raised
blister of
rust fungus



HOLLYHOCK RUST

Puccinia malvacearum

F: Pucciniaceae

This serious pest of hollyhocks is found in Eurasia and N. America. It covers the leaves with small pustules. Older leaves die and fall off.

PUCCINIA ALLII

F: Pucciniaceae

Common on onions, garlic, and leeks in Eurasia and N. America, this species forms pustules on infected leaves. These break open to release dust-like airborne spores.

rounded bright
orange-yellow pustules
erupt through leaf surface



powdery
pustules on leaf



HYPERICUM RUST

Melampsora hypericorum

F: Melampsoraceae

This common species from Eurasia is visible as scattered, raised pustules on the undersides of *Hypericum* leaves.

FUCHSIA RUST

Pucciniastrum epilobii

F: Pucciniastraceae

A parasite of fuchsias and fireweed in Eurasia, this species infects the leaves and forms pustules on the undersides of the leaves.